Foundational Models for Robotics

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1 Introduction

Building a general purpose robot, that is capable of behaving like a human functions has been the goal for a long time in Artificial Intelligence. The robot should perform a specific to general tasks, remain robust and adapt to different environments. This is a difficult task because any training which is required to build a robot, to perform a task required extensive labeled data which was time consuming to create and these robots eventually failed to generalize to real world scenarios. Recently, Foundational Models (LLMs, VLMs, ALMs) have gained a lot of traction in their ability to process natural language, perform common sense reasoning and interpret complex instructions given to them. The foundational models are good at these tasks because they have been trained on internet scale data which gives them the ability to perform in general situations. They are also flexible to be fine tuned further to perform down-stream tasks. These properties seem very suitable for the problems faced in robotics and in this technology review I focus on exploring the use of foundation models in the various aspects of robotics and how we can leverage them to build general purpose robots. This paper will be a good read for graduate students who are interested in the field of robotics and are curious to understand the impact the foundational models can have in this domain.

2 Background and Context

The domain of robotics has evolved from relying on deterministic frameworks to the development of sophisticated machines that show reasoning abilities and generalization of tasks that are comparable to human behavior. We need some context in the progression of robotics to understand the impact of foundational models in robotics. The overview is that every independent robotic system is mainly based on three autonomous systems. **Perception** is related to sensing and understanding the environment, **Decision Making** includes thinking and planning the actions and **Control** is ensuring that the action is implemented accurately and flexibly.

2.1 Robotics Before Machine Learning

The early robotics systems were deterministic, that is, they were designed to work in highly structured environments (Geiger et al., 2013). The autonomy stack before ML was simple but inflexible (Wood et al., 2018). Every layer of the operation required explicit programming. Perception relied on basic sensor data processing. Decision-making followed rule-based logic, and control executed pre-defined motions. For example, industrial robots like Unimate, used by car makers. Robots were able to execute motions more effectively and smoothly with the introduction of kinematic and dynamic models. Basic tasks, such as pick-and-place objects were doable using classical techniques in control like trajectory optimization. However, all these autonomy stacks of the robots were bounded by their inability in handling uncertainties. These uncertainties led to the shift from deterministic to probabilistic systems which used techniques like the Bayesian methods (Muehlemann et al., 2023) and Kalman filters for problems of localization and mapping, as were used in early SLAM implementations (Mur-Artal & Tardós, 2016). These systems were also based on human-defined features, and their decision-making processes were not robust to environmental changes.

2.2 Early Machine Learning in Robotics

With the popularity of machine learning during the 1990s, a different and data-driven approach was considered for the autonomy stacks in robotic systems. Since developing a robust set of rules for the policy of a robotic system proved to be challenging, Machine Learning techniques were being used to develop policies. One such popular technique was Learning from Demonstration (LfD) which allowed robots to learn tasks through human observation. Such policies were again only applicable to the states that were observed during the demonstrations and the corresponding actions that were taken. This means that the robotic system is still

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not robust to uncertain environments where the robot is in a state never encountered and trained for before. Another approach to obtain a policy is Reinforcement Learning. Here, the robot learned a policy based on its experiences to achieve a goal through rewards and penalties. RL increased the flexibility because robots could learn to extend policies through environmental interactions. However, there are a few key challenges such as complexity of high-dimensional continuous state and action spaces. More challenging is the problem of transferring the policies learnt to real-world environments, safety during learning, the sparseness of reward structures in many robotic tasks and the need for robustness with adaptability in dynamic environments. All of these indicated that more research was needed in the field.

2.3 The Deep Learning Revolution

The 2010s have seen the rise of deep learning revolutionize autonomy stacks with end-to-end learning capabilities. Perception layers became far stronger with CNNs, enabling real-time object detection (e.g., YOLO) and semantic segmentation (e.g., U-Net). These have enabled a few key tasks in autonomous navigation and manipulation with much better scene understanding. For example, SLAM systems evolved to incorporate DL-based perception for feature extraction and depth estimation. In decision-making, DL-powered reinforcement learning (Deep RL) transformed policy generation. Techniques like Deep Q-Networks (DQN) enabled robots to handle high-dimensional action spaces, as demonstrated by OpenAI's robotic hand solving a Rubik's Cube. Autonomous systems have increasingly relied on simulators, such as MuJoCo and Isaac Gym, to learn policies before transferring them to physical robots using Sim2Real methods. Earlier, the control layers were dominated by deterministic algorithms, they have been replaced by using neural networks for the adaptive execution of learned behaviors. However, despite these technological advancements, the autonomous systems remained limited by restrictions in computational efficiency and data requirements.

2.4 Foundational Models in Robotics

Foundation models, pre-trained on huge and very diverse datasets have achieved significant progress in vision and language processing. Robotics will also have promising applications of these models in the areas not only limited to autonomous driving, domestic assistance, industrial tasks, healthcare, field robotics, and multi-robot systems. Large Language Models, Vision-Language Models, and Audio-Language Models can be integrated in many robotics tasks, significantly enhancing perception, prediction, planning, and control abilities for robots. The rest of the review will be discussing the foundational models and their impact on robotics.

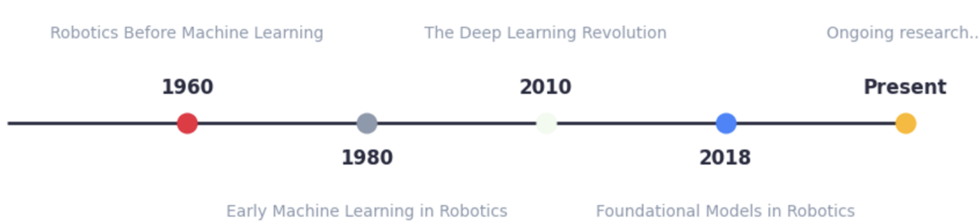

Fig. 1. Time line of key improvements in Robotics

The time line helps understand the stages of growth in robotics and where the foundational models make an impact.

3 Foundational Models

Foundational Models in Perception

Perception in robotics refers to the ability of robots to interpret sensory data from their environment, such as visual, tactile, or auditory inputs, to understand objects, scenes, and actions. Traditional perception methods

Table 1. Comparison of Methodologies

Methodology	Strengths	Limitations
Rule-Based Systems	Precise, deterministic	Lacked adaptability
Classical RL and LfD	Data-driven learning	Lacked robustness
Deep RL and LfD	Scalable, perception-aware policies	Poor generalization
Foundational Models	Generalization, multi-modal reasoning	Real Time Performance, Safety Evaluation

are largely based on task-specific models and hand-crafted features, which generalize poorly to diverse, unstructured environments or new situations. These methods also often require large quantities of labeled data such as demonstrations, they are sensitive to the presence of occlusions or noise, and lack the semantic understanding needed for higher-level reasoning. Without foundational models, robot systems face limitations in their ability to adapt to non pre-defined environments, interpret ambiguous language based instructions, or correctly combine the multi-modal sensor observations, which makes them less suitable for real-world applications under practical settings. Foundation models address these limitations. Their large-scale pre-training, open-vocabulary capabilities, and multi-task generalization enable more robust, adaptable, and intelligent robotic perception.

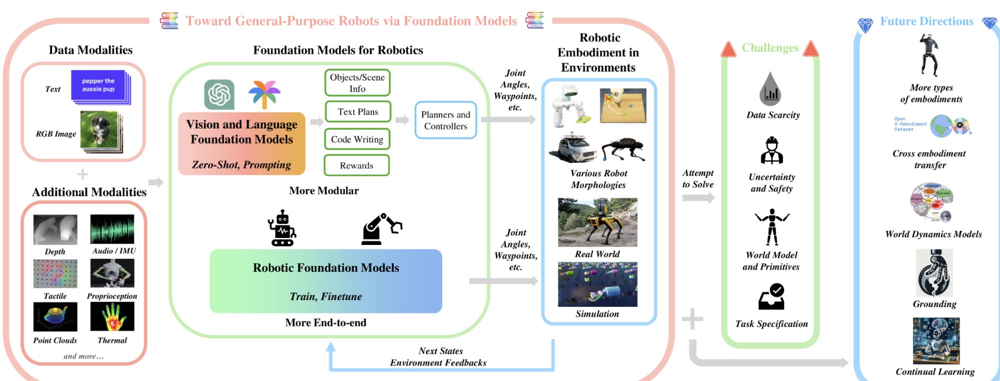

Fig. 2. Foundational Models in Robotics

3.1 Vision-Language Integration for Robotic Perception

CLIPort CLIPORT uses imitation-learning to combine the semantic understanding of CLIP and the spatial understanding of Transporter to perform tabletop manipulation tasks without explicit object representations or symbolic states. Integrated with pre-trained CLIP features, it achieves efficient few-shot learning and generalizes well to unseen tasks and concepts. The transporter shifts the objective of the model to detect actions instead of objects i.e. pick and place affordances. It supports multi-task and multi-goal policies for both simulated and real-world environments. This approach shifts from object-centric to action-centric perception, trying to adopt the strengths of vision-language models and overcome challenges in fine-grained manipulation with deformable and granular objects.

LM-Nav Using the foundation models LLM and VLM, LM-Nav relies on already trained models and does not require too much data. It contains a visual navigation model for mapping the robot's observations. The interpreting spoken or written instructions are broken down to generate pieces with an LLM and a VLM is used for grounding the textual landmarks to the robot's visual observations. and a LLM for interpreting spoken or written instructions and generating plans. The VNM then executes this plan. When these components are

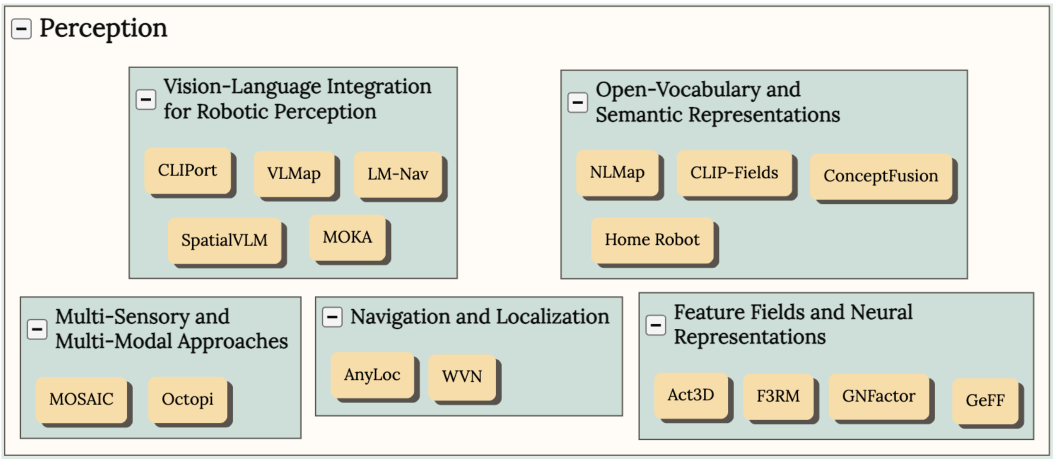

Fig. 3. Use Cases of Foundational Models in Robotic Perception
This figure highlights the different of foundational models in robotic perception, from CLIPort's imitation learning for tabletop manipulation to the integration of multimodal sensory data in MOSAIC and Octopi. Each model contributes uniquely to advancing robotic understanding, enabling more adaptive, flexible, and efficient robotic behaviors across different environments and tasks.

put together, LM-Nav can follow long-term, big-picture instructions in new places, adapt well, and carry out plans strongly without needing adjustments or specific human-labeled training data.

VLMap VLMaps combines visual based features from images with 3D representation of the physical world. Combined with LLMs, VLMaps can translate natural language instructions converted to a set of open-vocabulary objectives which are found in the map. The novel aspect of VLMaps is that they can identify spaces that aren't objects, e.g., "in "between the TV and the couch" beyond the capabilities some other models. It can develop new collision avoidance maps specific to the embodiments of the robot, e.g., "tables" are barriers to a tall mobile robot, but not for a drone.

SpatialVLM SpatialVLM is a system designed to improve how vision-language models (VLMs) understand space by creating and training on better spatial data. It produces large labeled datasets through open-vocabulary detection, depth estimation, image decomposition, and object focusing with captions. This data empowers VLMs to perform tasks like estimating distances from 2D images and correcting errors in current top VLMs like GPT-4V.

MOKA MOKA (Marking Open-world Keypoint Affordances) is an approach that allows the generalization of robots to novel tasks via simple natural language instructions with vision-language models (VLMs). It generates a minimal representation of affordances in terms of keypoints and waypoints over objects and tasks, bridging the gap between VLM predictions and robot movement. MOKA employs a step-by-step visual instruction to indicate what can be done with objects. It does this by using marks to direct the VLM's attention to important visual details. By choosing the best keypoints and waypoints in this manner, MOKA creates paths for different tasks like picking, placing, and using tools, requiring much less manual work and design help.

3.2 Open-Vocabulary and Semantic Representations

NLMap NLMap produces scene descriptions that are searchable in natural language, allowing for more accessible real-world planning. The system couples natural language commands with knowledge of the environment. It introduces a powerful framework for tasks like navigation and manipulation in complex and dynamic settings.

CLIP-Fields CLIP-Fields uses weakly supervised semantic fields for robotic memory and takes advantage of what CLIP can do. Mixing understanding of meanings with spatial details, CLIP-Fields thus helps robots create strong and lasting memory systems to understand and interact with their surroundings.

ConceptFusion ConceptFusion aims to create 3D maps that fuse types of information: semantics, appearance, and space. This method allows robots to build rich and adaptable maps of the surroundings, applicable in various tasks in unknown territories before.

HomeRobot HomeRobot is an open-vocabulary mobile manipulation system that combines object understanding with task execution. This system allows the robot to understand natural-language instructions for the execution of household tasks, such as organizing, cleaning, or assisting—showing flexibility and adaptability in the real world.

3.3 Feature Fields and Neural Representations

Act3D Key aspects of Act3D are it displays the robot's action space through a 3D field whose resolution is changed according to the task. The model moves up the 2D pre-trained features to 3D using depth sensing. It adopts a coarse-to-fine sampling strategy and utilizes relative-position attention to featurize 3D points. Act3D reaches state of the art on the RLBench manipulation benchmark and improves 10% absolutely over previous 2D multi-view policies.

F3RM F3RM (Feature Fields for Robotic Manipulation) focuses on few-shot language-guided manipulation using distilled feature fields. It performs distillation of features from 2D foundation models into 3D feature fields. F3RM allows for few-shot, manipulation based on language instructions with generalization for different object appearances, poses, categories and shapes. The model maps the characteristics of 2D images to the 3D models necessary for robot tasks.

GNFactor GNFactor generalizes neural feature fields for multi-task real robot learning. It collaborates in enhancing a general neural field (GNF) for reconstruction and a transformer for decision-making. GNFactor utilizes a VLM to convert detailed semantic information into a deep 3D voxel representation. The model exhibits a strong generalization capability for both seen and unseen tasks in realistic scenarios.

GeFF GeFF (Generalizable Feature Fields) is learning useful features for mobile manipulation. It provides one solution to perceive the scene for both robot movement and manipulation in real-time. GeFF adopts a generative novel view synthesis to pretrain its model. It wire the scenes generated with language using CLIP feature distillation. The model is effective in open-vocabulary mobile manipulation for dynamic scenes and generalizes to open-set objects.

These models have shown a trend toward more flexible, easy-to-adapt, and meaningful 3D representations for robots. They adopt advanced methods such as transformer designs, taking important features from main models, and adjusting resolution to perform better in a wide variety of manipulation and navigation tasks.

3.4 Multi-Sensory and Multi-Modal Approaches

MOSAIC: Learning Object Properties from Multiple Sensory Modalities MOSAIC (multimodal object property learning with self-attention and interactive comprehension) is a framework to learn the combination of the various sensory property representations for robotics perception. The important functions in MOSAIC are to combine the visual, audio, and haptic sensory modalities for a comprehensive understanding of object properties. It is based on the versitality of the embedding vectors of LLMs whose knowledge can be distilled for other sensory inputs. It applies multimodal fusion based on self-attention to fuse features of different modalities. It is competitive on object classification and object-fetching tasks despite using relatively simple linear classifiers.

Octopi is a method to integrate visual, tactile and semantic language input. Octopi (Object Comprehension with Tactile Observations for Physical Intelligence) uses the senses of touch along with the visual and text inputs in order to perform a task. In particular, they introduce a new dataset PhysiCLeAR where the object properties also contain information about the hardness, roughness, and bumpiness which can improve the performance of the robot. It uses a CLIP-type tactile encoder and LLaMA-type language model which processes the tactile signals together with the language commands. It shows how to predict object attributes from touch and use the predictions in reasoning tasks about scenarios.

MOSAIC and Octopi are both milestones in the field of multi-sensory and multi-modal for robotic perception and reasoning. MOSAIC incorporates visual, audio and haptic information for a general understanding of object features whereas Octopi fuse tactile sensing with language models to perform physical reasoning tasks. Such approaches display the possibilities of exploiting different sensors and modalities to facilitate richer perceptions and well grounded reasoning for execution capabilities by robots.

3.5 Navigation and Localization

WVN WVN (Wild Visual Navigation) is a self supervised model which tries to predict the traversability in the natural environment like woods using vision. Typically tall grass, tiny branches are seen as obstacles and alternate paths are found for the robot traversable, although it is still possible to go through the them. In order to overcome this issue WVN uses the vision transformers like DINO-ViT to extract detailed features of the images. With a short human demonstration and the features extracted, an online learning is done which allows autonomous navigation as it can segment walkable ground in less than 5 minutes of training in the field. This was tested on ANYmal, but it is generalizable to all.

AnyLoc AnyLoc tries to give a generic solution for Visual Place Recognition that works well in a broad variety of environments without retraining or fine-tuning. It works in different environments, including cities, outside, inside, in the air, underwater, and underground. AnyLoc exploits general purpose feature representations from off the shelf self supervised models (DINOv2). It combines derived features with unsupervised feature aggregation methods like VLAD. The features are extracted from intermediate layers/blocks of the vision transformer architechture. This allows AnyLoc to reach performance up to 4 times better than current methods.

Both WVN and AnyLoc represent significant advances in robot navigation and place recognition by leveraging the foundation models in powerful feature extraction which can be adapted to specific tasks without a lot of fine-tuning and training. This allows both WVN and AnyLoc to generalize across different environments.

Foundation Models in Task Planning

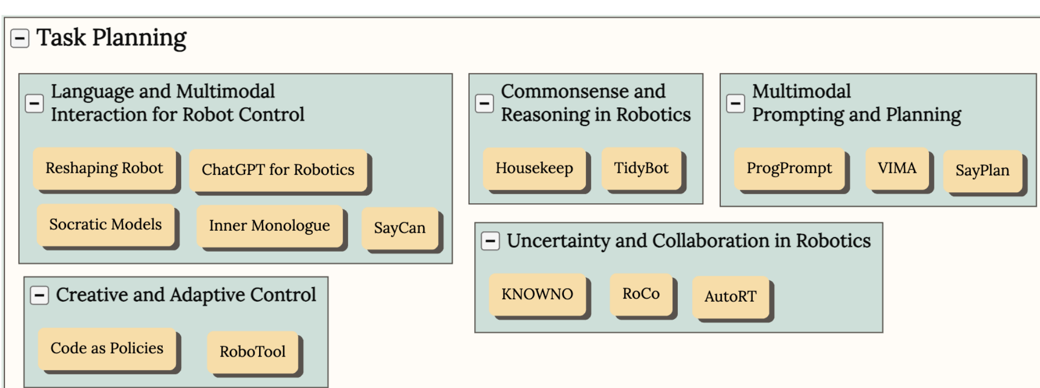

Fig. 4. Use Cases of Foundational Models in Robotic Task Planning

3.6 Language and Multimodal Interaction for Robot Control

Reshaping Robot Trajectories Using Natural Language Commands Language was not used much when humans wanted to communicate with the robot until recently. They always communicated with a rigid templates. This work provides a natural language based way to communicate with robots which lets a user alter the robots trajectory to perform an action. This solution uses LLMs (BERT and CLIP), where BERT is used to understand the commands and CLIP is used to associate the commands with the visual perception of the robot and combines these features with the current path information using using multi modal transformers.

It then treats the generation of the trajectory as a sequence prediction problem and uses imitation learning with a set of robot paths altered by language commands to adjust the robot trajectory. The model showed significant performance increase over baseline approaches in simulated trajectory scenarios. In experiments with a robot arm, users preferred this natural language interface over traditional methods.

Socratic Models: Composing Zero-Shot Multimodal Reasoning with Language Socratic Models aim to combine the different features of the foundation models (e.g., GPT-3, ViLD) and language-conditioned policies, using language as the common interface. This gives a mutually beneficial perfomance for all the combined models improving the robot and human interaction. For example, when a robot is given an instruction, it can parse the textual command using LLM better when it can also understand the visual context with VLM. This improves the reasoning and helps the robot interact with the human asking questions to form a clearer plan. Human: "Pour coffee in a cup on the table." Robot: "Should I use the blue or the red cup?". This question can only be asked by the robot when both the LLM and the VLM are used simultaneously for planning, which is enabled by the Socratic Model.

Correcting Robot Plans with Natural Language Feedback Current corrections involve real time human interaction or teleoperation in order to correct the robot plans. But the correction can be of any form from goal specification to introduction of new constraints. This system uses natural language to state corrections which can be accounted for in the cost function and therefore control the robot plans. It translates natural language inputs into transformations of cost functions, enabling robots to adapt their motions based on user preferences or to recover from planning errors. The two components involved are a motion planner which uses a predictive control to optimize paths by minimizing a specified cost and the language-parameterized cost correction module which processes environmental observations, the robot's state, and user-provided natural language corrections to produce updated cost functions. This approach achieves an $81\% - 93\%$ success rate in tasks where initial planning failed even after one or two corrections.

SayCan SayCan leverages LLM in order to bridge the gap between high level semantic knowledge and real world task execution. When given a long horizon task the LLM breaks it down into multiple short instructions, but the LLM is constrained to provide these instructions only in the feasible and contextually appropriate way based on training done on robot capabilities. It therefore combines the high level knowledge of the LLM with low level skills that are possible, providing the necessary grounding to perform a real world task. Its principle is Do As I Can, Not As I Say.

Inner Monologue: Embodied Reasoning through Planning with Language Models Inner Monologue approaches giving feedback from the perspective of how humans think in their mind when they are performing a task. When given a long horizon task, say open the door, they have an inner monologue which says Find keys, try opening the door with one key, if it doesn't work try another till the door is opened. They get the feedback through inner thinking like if it doesn't work try another key. Inner Monologue is a novel approach which does dynamic planning with iterative feedback using LLM planners for long horizon tasks. It essentially allows robots to follow a high-level natural language command by forming a plan by searching over a pre-trained short-horizon policies and iteratively improving plans through different types of feedback. This feedback is again given to the LLM as prompts allowing it to adapt to complete the long horizon task.

ChatGPT for Robotics The main goal of this work is to use ChatGPT to potentially have a non technical engineer who can interact with the robot and make it perform a task. For achieving this they have a high level robot function library. This library can be for a particular form factor or scenario, and it should correspond to implementations on the robot platform with clear documentation for ChatGPT to follow. Then when a prompt is created for ChatGPT that states what the objective is, it uses engineering design principle to provide an implementation using the library defined. The user remains on the loop for assessing code response from ChatGPT either by direct estimation or by simulations and gives feedback which ChatGPT uses to refine the code. The final code is then deployed on the robot. This approach is more from interaction between humans and robots from a development perspective which enables faster prototyping and testing.

3.7 Commonsense and Reasoning in Robotics

Housekeep Embodied agents lack the ability to tidy houses based on human preferences without explicit instructions. This work creates a dataset of human preferences for object placements in tidy and untidy houses called the Housekeep. Using LLMs embodied commonsense is extracted which is used to give a modular baseline. Fine-tuning the LLM on human preferences generalizes well and gives good performance when place in a new house with unseen objects.

TidyBot The TidyBot also attempts to tidy up rooms based on human preferences. Each person may have a different preference and the robot is able to quickly learn these preferences and perform the task effectively. It uses the summarization ability of LLMs to form a generalized rules based on the human preferences given as few shot prompts. For example, the prompts may be like "Yellow shirts go in the drawer" and "Purple shirts go in the shelf". The LLM is used to generalise this as brightly colored clothes go in drawers and dull colored go in shelves which is then applied in future scenarios when tidying up the room. It achieves 91.2% accuracy on unseen objects in the benchmark dataset and successfully tidies up 85.0% of objects in real-world test scenarios.

Both approaches demonstrate the potential of using foundation models to enhance commonsense reasoning and personalization in robotic household tasks, addressing the challenge of understanding and applying human preferences in complex environments.

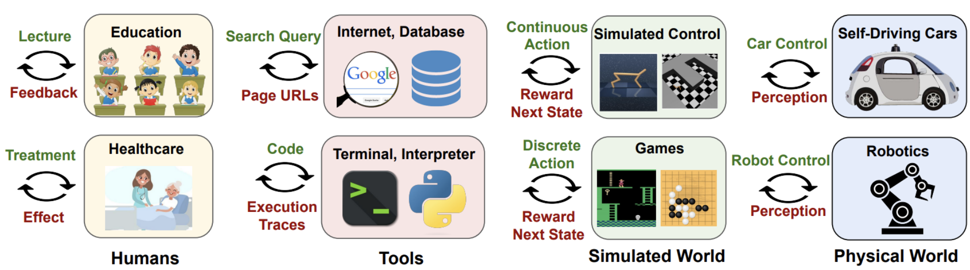

Fig. 5. Foundational Models in Robotics

3.8 Multimodal Prompting and Planning

VIMA VIMA (Visual-language Manipulation) enables to perform various robot manipulation tasks using prompts which can be text, images or videos. This multimodal prompting is a novel approach and the interleaving text and images/video is encoded as an input sequence for a transformer-based pre-trained LLM and the output is used to control the robot arm to accomplish the task. This gives the ability to learn from one shot prompts where videos are given as demonstrations or goal images can be used to understand the task. This gives a very good scalability as it uses the same architecture used for NLP tasks and outperforms alternate designs with 2.9x success rate.

ProgPrompt The use of LLMs to enable the robot to understand natural language instruction is very good, but the problem is that sometimes these instructions are not completely transferrable to actions because there is not enough information about the next actions available in a particular context which would enable to robot to understand if it is possible to perform the action. For example, when an instruction like "Put salmon in the fridge" is told to the robot, it may have not be near the salmon to put it in the fridge. Progmattic LLM prompt allows plan generation based on the situation of the robot with information about the actions possible and objects in an environment. It generates programs with assertions of whether an action is possible enabling efficient task planning thereby leveraging LLMs for both world knowledge and programming language understanding.

SayPlan SayPlan leverages LLMs for scalable task planning in robotics, specifically designed to address the challenges of grounding plans in large, multi-floor, and multi-room environments. It utilizes 3D Scene Graphs (3DSGs) to represent complex environments efficiently. When given a task the LLM conducts a search in the scene graph to identify the sub graph relevant to the task. This sub graph is explored to generate a high level task plan and a classical path planner completes the navigation path needed. The scene graph simulator gives feedback which is iteratively used by the LLM to finalise the plan avoiding infeasible actions. Exploring task relevant sub-graphs by LLM is very crucial since the large amount of information of multi floors cannot be comprehended effectively when an LLM that is not pre-trained is used.

3.9 Creative and Adaptive Control

Code as Policies Code as Policies employs Large Language Models (LLMs) in policy code generation. LLMs trained on code-completion are used to write robot policy code from natural language commands, allowing complex control policies to be created without manual programming. It uses few-shot prompting to guide the LLM in generating new policy codes for commands that it has never encountered before through independent combinations of the API calls. The chaining of traditional logic structures with references to third-party libraries can be used in creating LLM-generated code performing spatial-geometric reasoning for robotic tasks. It may comprehend such vague descriptions as "faster" and automatically assign specific values depending on the situation—a very common, everyday-sense behavior. This enables creating and adapting to different policies.

RoboTool RoboTool uses Large Language Models (LLMs) to help with creative tool use in robotics. It has an "Analyzer" that relies on LLMs for natural language instruction interpretation, capturing key task-related concepts. This enables more accessible human-robot interaction. The "Planner" component uses LLMs to create detailed plans from the language input and the crucial ideas uncovered by the Analyzer. This facilitates complex planning, which can include creative tool use. The "Calculator" component calculates appropriate parameters for each skill, which expands high-level plans into specific, executable actions. The "Coder" uses LLMs to transform the resulting plans into executable Python code. This helps connect ideas on planning with the actual control of the robot. RoboTool uses LLMs in this brain-inspired modular design showing how language models can significantly enhance the creative problem-solving abilities of robots.

3.10 Uncertainty and Collaboration in Robotics

KNOWNO KNOWNO applies LLM to robot planning and addresses a fundamental challenge - uncertainty estimation. KNOWNO leverages LLM capability to plan step-by-step and understand common sense. The framework allows LLM-based planners to know when they don't have sufficient information and request help when needed. KNOWNO generalizes the prediction theory to provide statistical guarantees over task completion. While ensuring that jobs get done, KNOWNO aims to reduce the demand on human help in complex planning scenarios. KNOWNO has been tested in various robot configurations, from simulated to real. It can handle tasks that may involve different types of ambiguity. It has demonstrated good performance compared to the current standard, including those using groups of models or detailed prompt adjustments, at improving efficiency and independence while offering formal guarantees.

RoCo RoCo finds new ways of using LLMs to assist robots in collaboration. Each robot is represented as an LLM agent in a dialogue. LLMs let robots reason about tasks and think over strategies in natural language. This strategy allows for high interpretability by human supervision. Then the LLMs make small task plans for every robot agent. These plans are continuously improved using feedback from the environment and checks. Given the start, goal, and obstacle locations, LLMs construct paths, waypoints, that incorporate crucial task information and environmental constraints. The waypoints generated by the LLM serve to accelerate and inform a multi-arm motion planner. The dialog-based setup allows for easy incorporation of human input. Users can converse and collaborate with robot assistants to complete tasks. This approach shows that LLMs can be more versatile in robotics, expanded from natural language processing to spatial reasoning and adaptive planning for multi-robot settings.

AutoRT AutoRT is a new system that adapts basic models for robots to work and collect data in new situations, largely beyond the help of people. Two are the major ways in which basic models are used in the system. VLMs are used for scene understanding and grounding, allowing robots to interpret their environment effectively. LLMs are applied to generate new instructions to be followed by a group of robots. This creates room for coming up with so many tasks grounded in the physical world. AutoRT can reason about the autonomy and safety tradeoffs of data collection leveraging FM. The system can instruct over 20 robots in various buildings to collect 77,000 robot episodes based on teleoperations and robot policy. The data that AutoRT gathers from real-life situations is much more diverse than regular datasets, which helps robots learn better. AutoRT can leverage LLMs to create robots that follow instructions to collect data which is aligned with human preferences, increasing the relevance and applicability of the data collected.

Foundational Models in Action Generation

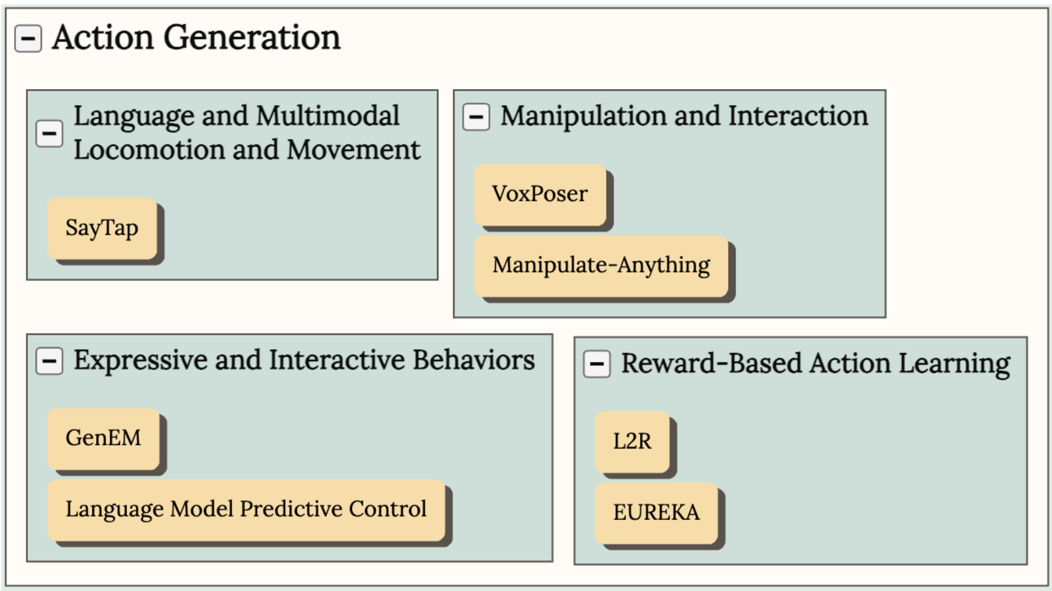

Fig. 6. Use Cases of Foundational Models in Robotic Action Generation

3.11 Locomotion and Movement

SayTap SayTap utilises LLMs in a creative way to connect simple natural language commands with detailed control over quadrupedal robots. LLMs, which assist in the comprehension and processing of natural language instructions by users, facilitate human-robot interaction. It uses a special prompt of a large language model, which converts natural language instructions into foot contact patterns. This acts as an intermediate layer between high level commands and low-level control actions. SayTap harnesses the broad knowledge of LLMs to synthesize a wide variety of movement behaviors from diverse natural language inputs. The researchers developed a particular method for designing prompts, which empowers the LLM to generate contact patterns suitable for the instructions. Such an LLM-based system can make users easily control the quadrupedal robot so that they can generate different walking styles dynamically. SayTap uses general knowledge captured in LLMs to guide generating complex contact patterns, effectively transferring that knowledge to the robotic locomotion domain. This approach demonstrates how the foundational models can be applied effectively to create simpler, more flexible means of controlling complex robots, thereby bridging the gap between what humans desire and what robots do.

3.12 Manipulation and Interaction

VoxPoser VoxPoser uses LLMs and VLMs, in new ways for robots to manipulate objects without limits. Free-form natural language instructions are interpreted by LLMs to infer the relevant affordances and constraints of the task at hand. LLMs generates Python code that interacts with perception APIs and manipulates 3D voxel maps for grounding abstract knowledge in the robot's observation space. VLMs like CLIP or open-vocabulary detectors are used to extract spatial-geometric information of relevant objects or parts in the environment. It composes dense 3D voxel maps, representing task-relevant rewards or costs in the observation space, from the LLM's inferences and VLM's visual grounding. VLMs provide generalizable visual grounding that allows the system to generalize to a large number of objects and environments. VoxPoser can create robot actions for many everyday tasks by instantiating these basis models in a planning system.

Manipulate-Anything Manipulate-Anything is an approach to automatically generate demonstration of real-world tasks for robot learning. It works in real-life situations without needing special state information or custom-made skills. It can transform any stationary object, not just specific ones learning from demonstrations on the fly. It works out a list of smaller goals and creates steps to achieve them. A checker checks the completion of sub-goals, which allows re-planning in case of a failure and alternate plans and also introduces recovery behavior into collected demonstrations. The system uses reasoning from different viewpoints to make VLM capabilities better and improve performance. Manipulate-Anything makes it easy to generate large amounts of high-quality, diverse robotic manipulation data. For example, when given the instruction “open the top drawer”, it can guide the robotic arm to complete the task based on various video demonstrations understood by the VLM.

3.13 Reward-Based Action Learning

L2R L2R (Language to Rewards) is a novel approach that uses LLMs to define reward parameters for robotic tasks. This closes the gap between high-level language instructions and low-level robot actions. LLMs are used to translate task semantics into reward functions, utilizing their code-writing capabilities. Rewards serve as an intermediate interface between high-level language instructions and low-level robot actions, addressing the challenge of hardware-dependent actions being underrepresented in LLM training data. The system combines LLM-generated rewards with MuJoCo MPC, a real-time optimizer, enabling immediate execution and feedback. Users can observe results in real-time and provide feedback, allowing for interactive steering of robot behavior. Reward terms are modular and compositional, enabling concise representations of complex behaviors, goals, and constraints. This approach allows non-technical users to generate and steer novel robot behaviors without extensive domain expertise or the need to engineer low-level primitives. The system was evaluated on 17 tasks using a simulated quadruped robot and a dexterous manipulator robot, achieving a 90% success rate compared to a 50% success rate for a baseline using primitive skills.

EUREKA EUREKA is a new algorithm that uses LLMs to generate reward functions, which create human-like reward functions for challenging robotic tasks. EUREKA utilizes cutting-edge LLMs like GPT-4, to construct rewards, write code, and make improvements with zero examples. It improves reward code gradually. Without prompting for specific tasks or reward templates pre-defined, EUREKA outperforms expert human-engineered rewards in 83% of tasks across 29 open-source RL environments with an average normalized improvement of 52%. The algorithm is shown to be effective on 10 different robot morphologies, including quadrupeds, quad-copters, bipeds, manipulators, and dexterous hands. EUREKA helps solve difficult tasks that were hard to do before using manual reward engineering. For example, it can spin a pen quickly with a simulated Shadow Hand. This method makes possible a new way to learn without using gradients for reinforcement learning from human feedback (RLHF). EUREKA can construct higher-quality, reward functions that are more human-aligned from many forms of human input, such as existing human reward functions and even simple text feedback. These findings show that EUREKA can connect advanced planning and complex robotics tasks and, therefore, provides a powerful tool for constructing automated reward functions in reinforcement learning.

3.14 Expressive and Interactive Behaviors

Generative Expressive Robot Behaviors using LLM This approach generates expressive robot actions using LLMs. Generative Expressive Motion (GenEM) is an approach for using LLMs to generate expressive behaviors for robots to interact with humans. This approach leverages the powerful social context in LLMs to induce appropriate expressive behaviors depending on the situation. GenEM uses few-shot prompting methods to convert human language instructions into control code with parameters by leveraging the known and learned skills of the robot. It can create actions using multiple capabilities of the robot including speech, body movement, and visual elements like light strips. GenEM can react to real-time human input. It then lets the system adjust and generate new expressive behaviors through recombining ones that it already possesses. LLMs help to create code that drives the robot, allowing it to perform complicated behaviors without recourse to rigid rules or hand-engineered data sets.

Language Model Predictive Control Language Model Predictive Control (LMPC) is an approach for using LLMs in the control of robots through extended interaction. LMPC fine tunes robot code-writing LLMs to remember in-context interactions and improve teachability. The human-robot interaction is formulated as a partially observable MDP, where the observations are the human language inputs, and the actions are the robot code outputs. Training an LLM to complete past interactions is considered training a transition dynamics model. The trained model is integrated with model predictive control to find shorter routes to success. LMPC helps non-expert teachers to perform better on new tasks by 26.9%. It also decreases the average number of human corrections from 2.4 to 1.9 over 78 tasks and 5 types of robots. LMPC demonstrates significant enhancement in LLM-based robot control for more efficient and personalized human-robot interaction over a longer period.

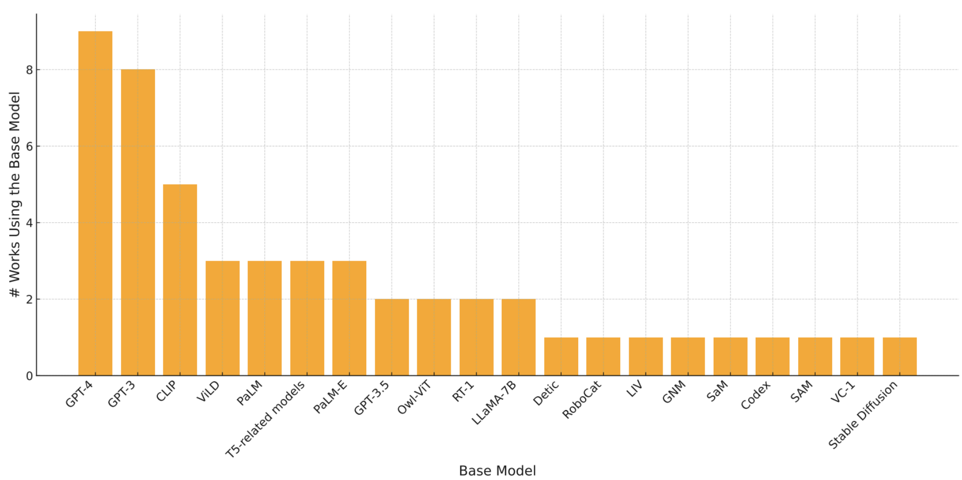

Fig. 7. Statistics of base models used for robotics

4 Robotics Foundational Models

So far we have explored how existing foundational models can be leveraged for robotics. The foundational models are trained on internet scaled data and are leveraged to perform robotics task. Robotics Foundation Models (RFMs) on the other hand are a class of AI models designed to perform robotics tasks. They are trained on large amounts of data from real robots, including actions and outcomes. These models make use of the increasing amount of robotics training data that can be used for a variety of robotic tasks. RFMs are of two main types, the Single-purpose RFMs and the General-purpose RFMs. The Single-purpose RFMs focus on one robotic capability, such as perception, planning, or control. Examples include, Action Generation Models which take simple sensory input, such as images or video, and generate control signals for parts of the robot, for example, RT series, RoboCat, and MOO. Motion Planning Models are concerned with the prediction of high-level motion-planning actions, mostly in visual navigation tasks. They often use rough topological maps instead of accurate metric maps. The General-purpose RFMs can do many tasks with different robotic parts, such as seeing, controlling, and even tasks that don't involve robots. Examples are Gato and PaLM-E. The RFMs learn in different ways from Imitation Learning, Reinforcement Learning at Scale, Vision and Language Pre-training, Self-supervised pre-training. The development of RFMs shows a huge change in robotics research, focusing on systems that can be used in many different ways. Important benefits of RFMs are better understanding across different tasks and forms, ability to leverage large-scale, diverse datasets, possibility for multimodal learning using different types of sensory inputs. RFMs are yielding good results in some areas, such as manipulation, navigation, and multitask learning, although they are still under development. The challenge remains to make these models work in more complex tasks and environments, connecting what we see in simulations to real life. We can only expect better and more advanced RFMs that will improve what robots can do as research progresses.

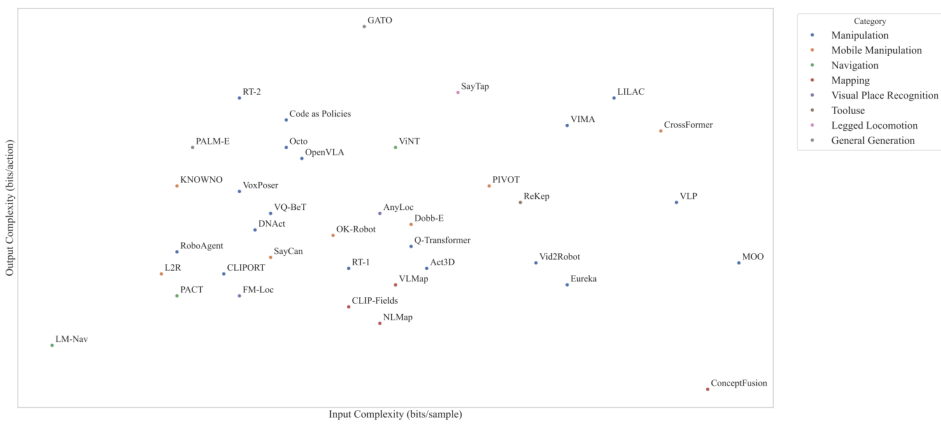

Fig. 8. Foundational Models in Robotics and Robotics Foundational Models
This figure shows the foundation models used in robotics and robotic foundation models plotted with respect to complexity of input data and complexity of output actions. The complexity of the input data comes from different types, such as image data being much more complicated than text data. The complexity of the output action space is mainly determined by the basic model output size in the robot tasks.

5 Discussions and Future Directions

Safety Safety and uncertainty quantification become very essential when using robots and foundation models in real-life scenarios, where they are collaborating with humans. It's vital to make sure such robots are safe and reliable because they're increasingly ubiquitous, spreading to factories, healthcare settings, homes, and offices. The use of foundation models in robotics has added a whole new level of possibility and challenge in this space. Older robot safety regulations include, hardware and software safety checks, collision avoidance systems and segregation of robots from workplaces for humans. But with the increased use of collaborative robots and AI-powered systems, new approaches to safety are needed. Lyapunov-style safety index functions provide hard safety for nonlinear systems with complex dynamics. Hamilton-Jacobi Reachability analysis is a concept that has been generalized to high-dimensional systems, e.g., quadcopter control. Safety-rule Reinforcement Learning where RL is combined with reachability analysis to learn safety value functions from complex inputs. Multi-agent safety formalizing safety for multi-agent interactions e.g., in urban autonomous driving scenarios. While these methods are promising, there are many challenges for their integration with robotic base models. Modeling safety as an affordance that varies depending on the robot's ability and social context. Making sure the robot knows what task it has to do and what the human user actually means. Foundation models currently are not capable of naturally understanding the uncertainty associated with their outputs. Proper measurement of uncertainty is important for activating backup safety plans, stopping actions that might be unsafe early, starting timed secure actions, requesting human-in-the-loop interventions. Some recent approaches to improve uncertainty quantification in robotics foundation models include conformal prediction, explicit constraint checking and ensemble disagreement. To improve safety and understanding of uncertainty in robotic foundation models, research should concentrate on establishing safer and more flexible safety rules that could be used in many places and by many professions, improving the accuracy of uncertainty estimates in foundation models. Combining primitive reasoning and common-sense knowledge from foundation models to help make safe decisions would be a good area to focus on.

Grounding Robot embodiment grounding is a hard problem that needs to deal with several major issues in an attempt to make better, more adaptive robotic systems. Although there have been many attempts to include grounding methods, these methods have some limitations and issues to address. Current approaches to connecting ideas to robot behaviors are highly limited. Natural language and code interfaces are poorly suited for complex actions, such as precise body movements. Predefined skill libraries are time-consuming to develop and struggle with generalization to new environments. Reward-based interfaces hold the promise in simulation but are confronted with safety concerns and time constraints in real-world applications. Going beyond a uni-modal grounding is an absolute necessity if the environment were to be known properly. Certain physical properties cannot only be captured through visual data alone. Interactive data with body feedback helps with concepts like friction. Other sensory modalities (e.g., olfaction) contribute to the complete sensory experience. Embodiment-Specific Grounding is also essential. Good grounding involves thinking about how the robot looks. Different types of robots (like humanoid and four-legged robots) need different action plans to do the same job. However, the current scholarship often favors environmental adaptation over embodiment-specific interaction strategies. To overcome these issues, future research in grounding robot embodiment is required.

Changing Environments Continual learning in robotics and artificial intelligence means a system's ability to learn and improve continuously. As soon as it knows new data or tasks, such a system must be capable of learning constantly. Continuing the learning process is very important while creating systems that are going to act in changing circumstances. It poses big challenges, with a problem that is popularly known as catastrophic forgetting. After training on new tasks, information previously learned by a neural network gets overwritten. Catastrophic forgetting is when Neural networks, when trained on new data, often forget previously learned tasks. This happens because of task interference and evolving data distributions. Retraining models from scratch on full datasets to avoid forgetting is both computationally costly and memory-intensive, making it infeasible for large models, so more efficient learning strategies become necessary. To solve these problems, some plans have been suggested, regularization techniques such as Elastic Weight Consolidation (EWC) and synaptic intelligence (SI) that try to preserve the important weights of the previous tasks. These are the ones involving replaying old data besides new data during training. Approaches like generative replay construct synthetic samples of the old data to mix with the new data. Progressive neural networks and dynamic expandable networks add new layers or units for each task, thus reducing overlap between tasks. Combining multiple strategies, such as replay with regularization, offers a balanced approach to maintaining old knowledge while learning new tasks. For robotics, continual learning is important for using robots in different settings. But, this field is not studied enough, especially when it comes to large language models. Important areas for research are designing strategies that properly mix the prior distribution of data with new data during fine-tuning to mitigate forgetting and improving the stability and efficiency of online learning algorithms for real-time applications. Continual learning is both a challenging and important area for the betterment of AI and robots. It needs new ideas to ensure that the computer works and performs strongly under changing tasks and situations.

Standardization and reproducibility are important problems in robotics research that need to be solved in order to improve the field. The robotics community realizes it is important to come up with practices that will ensure the possibility of checking and comparing published results across different studies. An important area of reproducibility is making the connection from simulated environments to real hardware. HomeRobot is a promising attempt to make both simulation and hardware platforms available for open-vocabulary pick-and-place tasks. It creates a common framework to test algorithms in simulation and real world. Developing standardized task definitions and affordances that can be used to treat different robot morphologies will allow for more efficient model development and comparison across studies.

Real World Scenarios The foundation models of robots can greatly change messy industrial places like construction sites. These workplaces are especially challenging because they are always changing, very large, and lack maps or outside communication. Further, dangerous materials, physical limits, and poor visibility create big problems in the way humans and robots can work together. Foundation models could change industries where robots are enabled to seek and explore challenging outdoor environments, do automated data analysis, generate automatic documentation, detect safety violations. There are some unique challenges to using foundation models in industry. Safety and risk awareness as the robots have to conform to stringent industrial safety norms. Industrial robots shall adapt to dynamic obstacles, varied surfaces, uneven terrains, and environmental conditions and the robot should be aware of possible hazards, such as heavy machinery. Deployment is needed at the edge to overcome network limitations and potential latency while on-site. To solve these problems, researchers are looking for alternative solutions which is the future areas of research. The foundation models of robotics can be significantly expanded to provide benefits in safety, efficiency, and real-time decision-making in harsh industrial environments as research progresses.

With the large number of researchers working on this field and with more gaps in the research areas, I think the development in robotics is going to be exponential and I anticipate that in the next 10-20 years, the robots will become more and more ubiquitous in our everyday lives where they will also abide by the safety rules and regulations.

6 Conclusion

In this review I went over the challenges faced in developing a robot which is robust and self sufficient in real world scenarios. I discuss the major robotic challenges before foundation models and where the foundation model come in and which aspect are they mainly trying to mitigate. I've broken down the main autonomous systems, perception, task planning and action control, that are necessary for a fully functioning robot. I group the latest researches and models by use cases in each of these three systems and discuss the latest research in these areas leveraging the vision and language foundation models. In addition to foundational models repurposed for robotics, there are developments on robotics foundation models which are trained on large scale robot training data. In the end I have mentioned a few of the challenges even with the foundation models and where the future work would be focusing on to overcome these challenges.

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